

MID TERM EXAMINATION**B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)**

Fourth Semester, May, 2023

Paper Code: ECC-216

Subject: Electromagnetic Field Theory

Time: 1½ Hrs.

Max. Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1 (a)	Calculate the area of the surface defined by $\rho = 10$, $\frac{\pi}{4} < \phi < \frac{\pi}{2}$, $0 < Z < 2$.	2.5	CO1
1 (b)	Find the electric field due to an infinite line of uniform charge ρ_L C/m lies along the Z axis.	2.5	CO1
1 (c)	Explain Gauss Law for magneto static field and its physical significance.	2.5	CO2
1 (d)	Compare the properties of different medium like free space, lossy dielectric, lossless dielectric, and perfect conductor	2.5	CO3
2 (a)	Explain the continuity of current equation and find the relaxation time in terms of permittivity and conductivity.	5	CO2
2 (b)	Two extensive homogeneous isotropic dielectrics meet on plane $Z = 0$. For $Z > 0$, $\epsilon_{r1} = 4$ and $Z < 0$, $\epsilon_{r2} = 3$. A uniform electric field $E_1 = 5 \hat{a}_x - 2 \hat{a}_y + 3 \hat{a}_z$ kV/m exists for $Z \geq 0$. Find (i) E_2 for $Z \leq 0$ (ii) The angles E_1 and E_2 make with the interface.	5	CO1
3 (a)	Find (i) ∇V , Where $V = e^{-z} \sin 2x \cosh y$ (ii) $\nabla \cdot \vec{A}$, Where $\vec{A} = \rho \sin \phi \hat{a}_\rho + \rho^2 z \hat{a}_\phi + z \cos \phi \hat{a}_z$ (iii) $\nabla \times \vec{A}$, Where $\vec{A} = yz \hat{a}_x + 4xy \hat{a}_y + y \hat{a}_z$ at $(1, -2, 3)$	1+2+2	CO1
3 (b)	Given $V = x^3 y^2 z$ and $\epsilon = 2.25 \epsilon_0$. Find (i) E at point P(-3, 1, 2) (ii) ρ_v at P.	5	CO1
4 (a)	What is the inconsistency of Ampere's Law and how it can be removed? Explain briefly.	5	CO3
4 (b)	Explain the magnetic boundary condition between two different dielectric media.	5	CO3